

FIRE-X

Remote inspection rover — Final components and budget

This is the final component list for the FIRE-X rescue rover. It groups all 49 budget lines into nine subsystems, each with a photo, quantity and price where one applies. Items (E) have no single named product and are planning estimates rather than supplier quotes; (R) and (I) items cost nothing extra.

1. Computer and control

Photo	Component	Qty	Unit ₹	Line ₹	Type
	01. Raspberry Pi Zero 2 W	1	₹2,210	₹2,210	L
	02. Raspberry Pi Pico H controller	1	₹739	₹739	L
	03. SanDisk 64 GB microSD card	1	₹1,889	₹1,889	L
	04. Pi heatsink, fan and mounting case	1	₹450	₹450	E
	05. Powered USB hub, OTG and data cables	1	₹650	₹650	E
	06. Interface PCB, headers and level shifting	1	₹700	₹700	E
	Subtotal			₹6,638	

2. Cameras and navigation

Photo	Component	Qty	Unit ₹	Line ₹	Type
	07. OV5647 5 MP Pi camera + cable	1	₹295	₹295	L
	08. ESP32-CAM with OV3660	1	₹689	₹689	L
	09. 7Semi MLX90640, 55° thermal array	1	₹4,585	₹4,585	L
	10. GY-53 / VL53L0X short-range module	4	₹599	₹2,396	L

Photo	Component	Qty	Unit ₹	Line ₹	Type
	11. MPU-6050 accelerometer / gyro	1	₹154	₹154	L
	Subtotal			₹8,119	

3. Gas and temperature

Photo	Component	Qty	Unit ₹	Line ₹	Type
	12. DFRobot SEN0466 CO sensor	1	₹7,419	₹7,419	L
	13. DFRobot SEN0322 oxygen sensor	1	₹6,048	₹6,048	L
	14. MQ-2 gas / smoke module	1	₹96	₹96	L
	15. SmartElex MAX31856 breakout	1	₹499	₹499	L
	16. Mineral-insulated K-type probe	1	₹700	₹700	E
	17. DS18B20 temperature probes	2	₹63	₹126	L
	Subtotal			₹14,888	

4. Sensor interfaces

Photo	Component	Qty	Unit ₹	Line ₹	Type
	18. ADS1115 ADC module	1	₹142	₹142	L
	19. 7Semi TCA9548A I2C multiplexer	1	₹97	₹97	L
	20. ACS758 100 A Hall current module	1	₹850	₹850	E
	21. Shaft encoders and mounting magnets	2	₹600	₹1,200	E
	22. Bumper microswitches and brackets	2	₹150	₹300	E
	23. LED lights, buzzer and status lamps	1	₹450	₹450	E

Photo	Component	Qty	Unit ₹	Line ₹	Type
	Subtotal			₹3,039	

5. Communication and operator

Photo	Component	Qty	Unit ₹	Line ₹	Type
	24. XIAO ESP32S3 + Wio-SX1262 kit	2	₹1,390	₹2,780	L
	25. LoRa external antennas and pigtails	2	₹300	₹600	E
	26. TP-Link TL-WR840N router + adapter	1	₹1,500	₹1,500	E
	27. USB gamepad with hold-to-drive button	1	₹700	₹700	E
	28. Existing laptop and charger	1	₹0	₹0	R
	29. Built-in Pi Wi-Fi and antenna	1	₹0	₹0	I
	Subtotal			₹5,580	

6. Drive system

Photo	Component	Qty	Unit ₹	Line ₹	Type
	30. MY1016Z2, 24 V 250 W geared motor	2	₹2,280	₹4,560	L
	31. Cytron MDDS30 dual motor driver	1	₹8,985	₹8,985	L
	32. Chain reduction, sprockets and guards	2	₹1,500	₹3,000	E
	33. Rubber track pair with drive sprockets	1	₹7,000	₹7,000	E
	34. Shafts, bearings, rollers and tensioners	1	₹3,500	₹3,500	E
	35. 6061 aluminium frame and	1	₹4,500	₹4,500	E
	Subtotal			₹31,545	

7. Battery and power

Photo	Component	Qty	Unit ₹	Line ₹	Type
	36. 12 V 12 Ah high-rate VRLA battery	2	₹2,000	₹4,000	E
	37. 24 V VRLA smart charger, 2 A	1	₹1,200	₹1,200	E

Photo	Component	Qty	Unit ₹	Line ₹	Type
	38. 7Semi AP64500, 5 V / 5 A converter	1	₹823	₹823	L
	39. 24-to-12 V buck converter, 3 A	1	₹450	₹450	E
	40. DC contactor, 24 V coil, 60 A or above	1	₹1,500	₹1,500	E
	41. Main/branch fuses, holders and isolator	1	₹800	₹800	E
	Subtotal			₹8,773	

8. Protection and assembly

Photo	Component	Qty	Unit ₹	Line ₹	Type
	42. NC mushroom E-stop and reset button	1	₹450	₹450	E
	43. Precharge, coil driver and surge network	1	₹650	₹650	E
	44. Harness, connectors and cable glands	1	₹1,800	₹1,800	E
	45. 304 stainless heat-shield panels	1	₹2,000	₹2,000	E
	46. Encapsulated thermal barrier + air gap	1	₹1,000	₹1,000	E
	47. Enclosures, mounts, fasteners and vents	1	₹1,100	₹1,100	E
	Subtotal			₹7,000	

9. Build and checking

Photo	Component	Qty	Unit ₹	Line ₹	Type
	48. Sensor check / calibration service	1	₹2,500	₹2,500	E
	49. Assembly consumables and spare parts	1	₹1,500	₹1,500	E
	Subtotal			₹4,000	

Cost summary

Component cost breakdown by subsystem

Subsystem	Amount ₹
Computer and control	₹6,638
Cameras and navigation	₹8,119
Gas and temperature	₹14,888
Sensor interfaces	₹3,039
Communication and operator	₹5,580
Drive system	₹31,545
Battery and power	₹8,773
Protection and assembly	₹7,000
Build and checking	₹4,000
Components and build costs	₹89,582

Final project cost

Total estimated project cost	₹90,000(estd.)
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